

VIGNESHWARAN N

palanivig12@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Results-driven Software Engineer with 2+ years of production experience designing scalable backend systems using Java, Microservices, Kafka, and REST APIs. Expertise in distributed systems, Design Patterns, and data structures and delivers 99.9% reliable solutions. Strong mentor who guides junior engineers through production debugging and facilitates technical knowledge-sharing in Agile environments.

TECHNICAL SKILLS

- **Languages & AI:** Java, Python, Go, SQL, AI & ML Concepts, Agentic AI, Model Context Protocol (MCP), Prompt Engineering, Data
- **Systems & Architecture:** Microservices, Distributed Systems, REST APIs, Message Queues (Kafka), Distributed File Systems (HDFS), Docker, Monitoring & Logging, Software Design Patterns, SDLC
- **Testing & Quality Assurance:** Unit Testing, Integration Testing, Debugging, Test Automation, Code Quality
- **Data & Cloud:** Relational Databases (PostgreSQL), Columnar Databases (ClickHouse), NoSQL (MongoDB), Cloud Queues (AWS SQS), Cloud Platforms (AWS), Git, Shell Scripting
- **Development Practices:** Agile/Scrum Methodologies, Problem-Solving, Continuous Integration, Automated Deployment Pipelines, Production Debugging, Security Best Practices

PROFESSIONAL EXPERIENCE

ZOHO – *Software Development Engineer*

May 2024 – Present

- Delivered **15+ production enhancements** for large-scale Java distributed systems, supporting mission-critical infrastructure serving **thousands of global users**.
- Engineered Kafka middleware for streaming pipelines, optimizing multithreaded data ingestion to scale throughput to **terabyte-level** volumes with **zero message lag**.
- Streamlined Linux-based CI/CD rolling upgrade pipelines via shell scripting, reducing manual operational overhead by **60%** and accelerating release cycles.
- Migrated core REST APIs to standardized client-server protocols, accelerating downstream integration times by **40%** for **10+ internal teams**.
- Architected an event-driven, multi-channel notification framework utilizing message queues and data stores, ensuring **99.9% reliability** through rigorous automated testing.
- Built continuous operational monitoring suites tracking **100+ queue instances**, demonstrating strong ownership and reducing issue detection time by **50%**.
- Designed encryption protocols and scalable data structures for a strategic infrastructure migration to ClickHouse, optimizing high-volume query performance and search availability.

ZOHO – *Software Development Engineer Intern*

Feb 2024 – Apr 2024

- Patched a critical multithreading concurrency bug in **Apache Tomcat's** caching mechanism, successfully merging the fix into the official global open-source repository.
- Hardened system security across multiple services by migrating hardcoded secrets to an encrypted, centralized decryption interface.
- Automated **10+ operational workflows** in Linux using shell and XML scripts, streamlining certificate management and reducing manual operations by **8+ hours weekly**.

OPEN SOURCE & ADDITIONAL HIGHLIGHTS

- **Open Source Contributions:** Official contributor to **Apache Tomcat** (Concurrency Bug Fix - Merged).
- **Technical Writing:** Created comprehensive documentation and facilitated technical knowledge-sharing sessions in Agile environments.
- **Continuous Learning:** Active research and prototyping in emerging technologies, specifically AI/ML integrations and distributed systems optimization.

CERTIFICATIONS

Introduction to Data Analytics (2022), Data Structures And Algorithms (2026), System Design Patterns (2026)

EDUCATION

Saranathan College of Engineering, Tiruchirappalli
B.E. Computer Science and Engineering

May 2024
GPA: **9.13/10**